

Resolution Geometry

The Theorem Ladder

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Abstract

Resolution Geometry (RG) is the geometry of what a declared family of observation protocols can distinguish. Its primitive is the equality-of-law relation of a joint observation law. The relation defines a kernel groupoid and a quotient; the stabilizer of the law modulo its gauge part defines an effective symmetry group acting on the quotient. On regular strata every structure computed from laws descends to the quotient and is invariant under that group. Differential, statistical, causal, temporal and metric structure are added in a fixed order, each either forced by the preceding objects, imported from established mathematics, or conditioned on a named hypothesis. This paper states the complete ladder from the exact kernel to calibrated physical distance, with proofs of the RG-specific results, explicit boundaries, and the remaining open problems.

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1 Conventions

Scope. Unless stated otherwise, latent spaces are finite-dimensional smooth manifolds, experiments are classical, dominated and regular in the sense of Section 5, and all statements are local to a stratum. Singular loci are treated by stratification.

Status of results. Each result carries one status.

- *Theorem, Proposition, Lemma*: proved here from the preceding objects.
- *Imported Theorem*: an established result, cited and not reproved.
- *Hypothesis*: a named condition H -...; a result that uses it states it.
- *No-Go*: a stronger statement is false; a counterexample is given.
- *Open Problem*: no result is asserted.

A result is called Γ -conditional when its hypothesis is that a stated subgroup belongs to the effective symmetry group of Definition 4.1. Such a hypothesis is tested by equality of laws.

Hypotheses used in this paper.

H-sound	declared groupoid lies in the kernel groupoid (Def. 4.5)
H-compl	declared groupoid generates every fibre (Def. 4.5)
H-coh	declared groupoid is normalized by the declared physical group (Prop. 4.8)
H-reg	regular stratum: submersive quotient, immersive descended law (Def. 5.1)
H-nuis	constant target and nuisance rank, or a declared nuisance bound (Hyp. 6.8)
H-sens	compatible sensor family (Def. 7.1)
H-top	topology and regularity for a cone boundary (Hyp. 9.2)
H-int	exact scale cocycle (Hyp. 10.1)
H-SI	SI metrological anchor (Hyp. 12.8)
H-med	propagation speed anchored or tomographically resolved (Hyp. 12.11)
H-val	held-out anchor validation (Hyp. 12.13)
H- Γ	a stated subgroup belongs to Γ (Def. 4.1)

2 Objects

Definition 2.1 (Observation system). An *observation system* is a triple $\mathcal{S} = (X, \{F_i\}_{i \in I}, \Psi)$ consisting of

1. a latent space X ;
2. protocols $F_i : X \rightarrow Y_i$, where Y_i is a space of observable laws (probability laws of records, or deterministic records);
3. a declared relabeling group $\Psi = \prod_i \Psi_i$, where Ψ_i acts on Y_i by pushforward along admissible bijections of outcomes.

The *joint law* is $F = (F_i)_{i \in I} : X \rightarrow Y = \prod_i Y_i$. A single experiment is the case $|I| = 1$.

Definition 2.2 (Counted channel). A protocol is *counted* if its records take values in a finite chain $\{0, 1, \dots, N\}$ or in $\mathbb{N}$, and its admissible relabelings preserve order.

Definition 2.3 (Natural module). A *natural module* is a pullback $M = F^* \mu$ of a Ψ -invariant structure μ on Y : a function, a tensor field, a cone field, a family of functions with transition maps, or a transport defined from laws. A structure on X that is not of this form is *latent*.

Example 2.4. Observable invariants, clock readings, record counts and the Fisher tensor are natural. A transport generated by a latent dynamical law is latent unless it is defined from laws.

3 The exact layer

Theorem 3.1 (Kernel pair and quotient). *The relation $x \sim x' \iff F(x) = F(x')$ is an equivalence relation. Its graph*

$$\mathcal{K} = X \times_Y X \rightrightarrows X, \quad s(x, x') = x, \quad t(x, x') = x', \quad (x', x'') \circ (x, x') = (x, x''),$$

is a groupoid, the kernel groupoid. The quotient $\pi : X \rightarrow Q = X/\mathcal{K}$ satisfies $Q \cong \text{im } F$ canonically.

Proof. Reflexivity, symmetry and transitivity are inherited from equality in Y , and these are the groupoid axioms for the graph. The map $[x] \mapsto F(x)$ is well defined and bijective onto $\text{im } F$. $\square$

Theorem 3.2 (Universal factorization). *A map $H : X \rightarrow Z$ is constant on $\mathcal{K}$ -orbits iff there is a unique $\bar{H} : Q \rightarrow Z$ with $H = \bar{H} \circ \pi$. In particular the observable invariants are exactly the functions on Q .*

Proof. Set $\bar{H}([x]) = H(x)$. It is well defined by fibre constancy and unique because π is surjective. $\square$

Proposition 3.3 (Joint kernel). $\mathcal{K} = \bigcap_i \mathcal{K}_{F_i}$. *The induced map $Q \rightarrow \prod_i X/\mathcal{K}_{F_i}$ is injective and in general not surjective.*

Proof. Equality in a product is equality in every factor. For non-surjectivity take $X = \{1, 2\}$, $F_1 = F_2 = \text{id}$: the product of quotients has four points, Q has two. $\square$

Proposition 3.4 (Garbling). *Let M be a Markov kernel independent of x . Then $\mathcal{K}_F \subseteq \mathcal{K}_{M \circ F}$, with strict inclusion in general.*

Proof. $P_x = P_{x'}$ implies $MP_x = MP_{x'}$. Merging two outcomes identifies $(0.1, 0.3, 0.6)$ with $(0.3, 0.1, 0.6)$. $\square$

No-Go 3.5 (Limits of the exact layer). The kernel groupoid and the quotient do not determine a topology, a differential structure, a metric, a statistical order, a reachability order, a cone, a scale, a connection, a clock, or a unit. Each of these is added in a later section.

4 The symmetry layer

Definition 4.1 (Stabilizer, gauge, effective symmetry).

$$\text{Stab} = \{(\varphi, \psi) \in \text{Diff}(X) \times \Psi : F \circ \varphi = \psi \circ F\}, \quad \text{Gau} = \{\varphi \in \text{Diff}(X) : F \circ \varphi = F\}.$$

Pairs whose relabelings agree on $\text{im } F$ are identified. The *effective symmetry group* is $\Gamma = \text{Stab}/\text{Gau}$.

Theorem 4.2 (Kernel exact sequence). 1. Every element of Stab maps $\mathcal{K}$ to itself: $(\varphi \times \varphi)(\mathcal{K}) = \mathcal{K}$.

2. Gau is a normal subgroup of Stab .

3. Γ acts faithfully on Q by $[\varphi] \cdot [x] = [\varphi x]$, and

$$1 \longrightarrow \text{Gau} \longrightarrow \text{Stab} \longrightarrow \Gamma \longrightarrow 1$$

is exact.

Proof. If $F(x) = F(x')$ then $F(\varphi x) = \psi F(x) = \psi F(x') = F(\varphi x')$; this gives (1) and a well-defined action on Q . An element acts trivially on Q iff $F \circ \varphi = F$, i.e. iff it lies in Gau ; this gives normality and faithfulness. $\square$

No-Go 4.3 (Non-splitting). The sequence of Theorem 4.2 need not split. For $X = \mathbb{Z}_4$, $F(x) = x \bmod 2$ and the declared group $\mathbb{Z}_4$ of rotations, the gauge part is $\{0, 2\}$ and the image in $\text{Aut}(Q)$ is $\mathbb{Z}_2$; the only subgroup of order two is the gauge part itself. An observable symmetry therefore need not lift to a latent symmetry of the same order.

Lemma 4.4 (Count rigidity). The order-preserving bijections of a finite chain are the identity. Hence Ψ_i is trivial on every counted channel, and no element of Stab rescales a counted record.

Proof. An order-preserving bijection of $\{0, \dots, N\}$ fixes 0 and, inductively, every successor. $\square$

Definition 4.5 (Declared assembly). A declared groupoid is a groupoid $A \rightrightarrows X$. It is *sound* (**H-sound**) if $A \subseteq \mathcal{K}$ and *complete* (**H-compl**) if $A \cdot x = [x]$ for all x . A declared physical group is a subgroup $G \leq \text{Stab}$ with image $\Gamma_G \leq \Gamma$.

Theorem 4.6 (Orbit-quotient comparison). Under **H-sound** the map $X/A \rightarrow Q$, $[x]_A \mapsto [x]$, is a well-defined surjection. It is a bijection iff **H-compl** holds.

Proof. Soundness gives well-definedness; surjectivity of π gives surjectivity; injectivity is the statement that each fibre is one A -orbit. $\square$

No-Go 4.7 (Saturation hierarchy). The following are distinct: (a) full differential rank of a family of A -invariants; (b) global separation of A -orbits by those invariants; (c) completeness; (d) reconstruction of a latent representative. The map $(u, v) \mapsto (u^2, v)$ has full generic rank and identifies $(\pm u, v)$; a coordinate absent from F and not moved by A produces several A -orbits in one fibre; an orbit does not select a representative.

Proposition 4.8 (Coherence). If $gAg^{-1} = A$ for all $g \in G$ (**H-coh**), the completeness locus $\{x : A \cdot x = [x]\}$ is G -invariant. If Γ_G acts transitively on a stratum, completeness at one point of the stratum implies completeness on the stratum. Without **H-coh** completeness can depend on the point: on $\mathbb{Z}_6 \times \mathbb{Z}_2$ with $F(x, b) = x$, bit flips declared at even x are sound and complete exactly at even x .

Proof. $g(A \cdot x) = (gAg^{-1})(gx) = A \cdot gx$, and $g[x] = [gx]$ by Theorem 4.2. $\square$

5 The differential layer

Definition 5.1 (Regular stratum). A regular stratum is an open $\mathcal{K}$ -saturated submanifold $S \subseteq X$ such that $Q_S = S/\mathcal{K}$ is a manifold, $\pi|_S$ is a submersion, and the descended law $\bar{F} : Q_S \rightarrow Y$ is an immersion into a smooth model of law space (**H-reg**). The observational vertical space is $V_x = \ker D\pi_x$.

Theorem 5.2 (Tangent sequence). *On a regular stratum*

$$0 \longrightarrow V \longrightarrow TS \xrightarrow{D\pi} \pi^* TQ_S \longrightarrow 0$$

is exact, so $T_{[x]}Q_S \cong T_x X/V_x$. For a sound declared groupoid, $V_x^A = T_x(A \cdot x) \subseteq V_x$, and V_x/V_x^A is the infinitesimal completeness defect.

Theorem 5.3 (Natural descent). *On a regular stratum let $M = F^*\mu$ be a natural module. Then*

1. $M = \pi^* \bar{M}$ with $\bar{M} = \bar{F}^* \mu$ unique; M annihilates V and is independent of the representative in each fibre;
2. $\bar{M}$ is Γ -invariant;
3. if M is a symmetric tensor and μ is nondegenerate, $\text{rad } M = V$.

Proof. (1) $F^*\mu = (\bar{F} \circ \pi)^*\mu = \pi^* \bar{F}^* \mu$. (2) For $(\varphi, \psi) \in \text{Stab}$, $\varphi^* F^* \mu = (\psi \circ F)^* \mu = F^* \psi^* \mu = F^* \mu$. (3) $\bar{F}$ is immersive and μ nondegenerate, so $\bar{M}$ is nondegenerate and the radical of $\pi^* \bar{M}$ is $\ker D\pi$. $\square$

Corollary 5.4 (Automatic descent). *For natural modules, basicness with respect to $\mathcal{K}$, representative independence, and projectability of law-defined transports and law-defined cone fields hold without further hypothesis. For latent structures they are hypotheses.*

Imported Theorem 5.5 (Singular strata; [12, 15, 23, 7]). *Orbit spaces of proper Lie groupoids carry orbit-type stratifications with slice models. At rank, isotropy or orbit-type changes the single tangent quotient is replaced by slice data and tangent cones.*

Proposition 5.6 (Tangent cone at an isotropy wall). *For $\mathbb{Z}_2$ acting by $(u, v) \mapsto (u, -v)$ and $\pi(u, v) = (u, v^2)$, $D\pi_{(0,0)}$ has rank one, while jet-rescaled secants $(\Delta u/h, \Delta r/h^2)$ fill $\mathbb{R} \times \mathbb{R}_{\geq 0}$. The first-order tangent misses a one-sided observable direction; higher jets of F are required at walls.*

6 The second-order layer

Definition 6.1 (Fisher module). *On a dominated regular experiment with score $S_u = \partial_u \log p(y | x)$, the Fisher tensor is $g_F(u, w) = \mathbb{E}[S_u S_w]$. It is the pullback of the Fisher–Rao metric along the law map and is a natural module.*

Proposition 6.2 (Fisher descent and ignorance). *On a regular stratum $g_F = \pi^* \bar{g}_F$, $\bar{g}_F$ is Γ -invariant, and $\text{rad } g_F = V$. For $Y = Jx + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, \Sigma)$, $\Sigma \succ 0$: $g_F = J^\top \Sigma^{-1} J$ and $\text{rad } g_F = \ker J$.*

Proof. Theorem 5.3 with μ the Fisher–Rao metric; the Gaussian formula is standard. $\square$

Imported Theorem 6.3 (Regular reconstruction; [19, 22]). *The resolved sector $R = \text{im } J$, the null sector $N = \ker J^*$ and the covariance block geometry are gauge-equivariant functions of (J, Σ, g_O) ; the Fisher tensor is the inverse Schur complement of the covariance on the resolved block.*

No-Go 6.4 (Fisher insufficiency). g_F does not determine the covariance. Covariances with equal Schur complement on the resolved block and different null blocks have the same Fisher tensor.

Proposition 6.5 (Second-order quotient; [18]). *Let $O(p) \times O(q)$ act on triples (Σ_R, Σ_N, C) . On the regular stratum the orbit space has dimension $p + q + pq$. The two spectra and the $\min(p, q)$ canonical correlations therefore do not coordinate it when $pq > \min(p, q)$; the mixed traces $\text{tr}(\Sigma_R^a C \Sigma_N^b C^\top)$ over finite exponent ranges supply generically independent coordinates, up to a finite ambiguity.*

Proof of the dimension. Ambient dimension $\frac{p(p+1)}{2} + \frac{q(q+1)}{2} + pq$ minus group dimension $\frac{p(p-1)}{2} + \frac{q(q-1)}{2}$, with finite generic stabilizer. $\square$

No-Go 6.6 (Shadow is not law). Distinct laws can share means and covariances (higher-moment escape), and processes with equal one-time covariance can differ in their two-time laws (memory escape). The second-order shadow is a natural module of the law, not the law.

Definition 6.7 (Efficient metric). For target coordinates q and nuisance coordinates ν , $g_{\text{eff}} = I_{qq} - I_{q\nu} I_{\nu\nu}^+ I_{\nu q}$.

Hypothesis 6.8 (H-nuis). Target and nuisance ranks are constant on the stratum, or a bound on the nuisance is declared and included in the efficient metric.

Proposition 6.9 (Nuisance conventions). g_{eff} is invariant under $\nu \mapsto \nu + f(q)$ and depends on the declared target chart. Without H-nuis it can be discontinuous: for $I_\varepsilon = \begin{pmatrix} a & \rho\sqrt{a\varepsilon} \\ \rho\sqrt{a\varepsilon} & \varepsilon \end{pmatrix}$, $\lim_{\varepsilon \rightarrow 0^+} g_{\text{eff}} = a(1 - \rho^2)$ whereas $g_{\text{eff}} = a$ at $\varepsilon = 0$.

Imported Theorem 6.10 (Data processing; [5, 1]). For a parameter-independent Markov kernel, $g_{F, M \circ F} \preceq g_{F, F}$.

No-Go 6.11 (Kernel order is not Blackwell order; [3, 11]). Two injective experiments have the same kernel (the diagonal) but need not be Blackwell equivalent: $\mathcal{N}(x, 1)$ and $\mathcal{N}(x, 4)$ are both injective and the second is a strict garbling of the first.

7 Ignorance and the observation limit

Definition 7.1 (Persistent null). For a family $\{F_\rho\}$ of protocols on a common latent space (H-sens), the persistent null at x is $\mathcal{I}_x = \bigcap_\rho \ker DF_\rho|_x$.

Proposition 7.2 (Finite attainment). In finite dimension $\mathcal{I}_x$ equals the intersection over finitely many members of the family, and it equals V_x of the joint law of that finite subfamily on its regular stratum.

Proof. A descending chain of subspaces of a finite-dimensional space stabilizes. The joint statement is Proposition 3.3 at the tangent level. $\square$

Proposition 7.3 (Three layers). For smooth maps E (evaluation at finite precision) and R (downstream representation), the exact null of the sensor is $\ker DF$. The precision loss of E and the representation loss of R are nulls of the composites $E \circ F$ and $R \circ E \circ F$ and satisfy $\ker DF \subseteq \ker D(E \circ F) \subseteq \ker D(R \circ E \circ F)$. Only $\ker DF$ is a property of the sensor.

Imported Theorem 7.4 (Finite-sample boundary reconstruction; [17]). For a correctly specified regular protocol with a design that places positive mass on both sides of an operational boundary and a nonsingular Fisher matrix, the maximum-likelihood boundary estimator is consistent and asymptotically normal with covariance given by the inverse Fisher matrix. The quotient alone determines no numerical boundary: affine recalibration of the latent coordinate moves it.

8 Statistical distance and angle

Definition 8.1 (RG length, distance, angle). On a regular stratum on which $\bar{g}_{\text{eff}}$ is positive definite,

$$L[\gamma] = \int \sqrt{\bar{g}_{\text{eff}}(\dot{\gamma}, \dot{\gamma})} dt, \quad d_{RG}(q_0, q_1) = \inf_\gamma L[\gamma], \quad \cos \angle(u, w) = \frac{\bar{g}_{\text{eff}}(u, w)}{\|u\| \|w\|}.$$

Theorem 8.2 (Distance law). *Under $H\text{-reg}$ and $H\text{-nuis}$, d_{RG} is the Riemannian distance of $\bar{g}_{\text{eff}}$ on Q_S . It is independent of the latent representative and of the latent chart, Γ -invariant, and dimensionless.*

Proof. Theorem 5.3 and Proposition 6.2; the Fisher tensor is invariant under reparametrization and the score is dimensionless. $\square$

Proposition 8.3 (Monotonicity). *Under a parameter-independent garbling, path lengths and d_{RG} do not increase.*

No-Go 8.4 (Rank one). On a rank-one stratum, length and order along the resolved direction are defined, and no angle between independent directions is.

No-Go 8.5 (Finite divergences). Only the intrinsic length is additive under concatenation. For the Gaussian location family the Bhattacharyya angle between p and $p(\cdot - a)$ is $2 \arccos e^{-a^2/8\sigma^2}$, which is not linear in a and is bounded by π .

No-Go 8.6 (Experiment relativity). For N samples of $\mathcal{N}(x, \sigma^2)$, $d_{RG} = \sqrt{N} |\Delta x| / \sigma$. On a class of experiments closed under additive-noise garbling there is no function f with $f(d_{RG}) = |\Delta x|$: $d_{RG}(\Delta x = 1, \tau = 0) = d_{RG}(\Delta x = 2, \tau = \sqrt{3}\sigma)$ for added noise of variance τ^2 .

9 Order, cone and conformal class

Definition 9.1 (Reachability). A declared preorder $\preceq$ on X is *fibre-compatible* if it is constant on $\mathcal{K}$ -orbits. It descends to Q ; its antisymmetrization is the causal order.

Hypothesis 9.2 ($H\text{-top}$). Q_S carries a topology compatible with $\preceq$, and the future sets have a regular boundary whose tangent directions form a cone field $C_q \subset T_q Q_S$.

No-Go 9.3 (Projective typing). The realization type of C_q is determined by its projective boundary, not by a defining function: $\{Q = 0\}$ and $\{Q^3 = 0\}$ define the same cone.

Theorem 9.4 (Realization branches). *Under $H\text{-top}$ the first applicable case describes C_q on a stratum:*

1. *the null cone of a nondegenerate Lorentzian quadratic form: then C determines a positive conformal ray $[g]$ (Imported Theorem 9.5);*
2. *a single smooth, strongly convex, homogeneous cone with Lorentzian fibre Hessian: a Lorentz-Finsler structure;*
3. *several persistent characteristic sheets: a multicone or multi-Finsler structure;*
4. *a cone whose rank, signature, topology, isotropy or sheet count changes: stratified realization by slices and tangent cones (Imported Theorem 5.5);*
5. *otherwise: general characteristic or admissibility geometry.*

Imported Theorem 9.5 (Conformal reconstruction; [8, 13]). *Two nondegenerate Lorentzian quadratic forms on a real vector space with the same null cone are proportional; a time orientation fixes the sign. Globally, under distinguishing causality, the causal structure of a Lorentzian manifold determines its conformal class and not its conformal factor.*

Theorem 9.6 (Lorentz isotropy forces the quadratic branch). *Suppose the isotropy Γ_q acts on $T_q Q \cong \mathbb{R}^{1+n}$ and contains $SO^+(1, n)$ in its standard representation, $n \geq 2$ ($H\text{-}\Gamma$). Then every Γ_q -invariant closed convex pointed cone with nonempty interior is the future or the past light cone, and the Γ_q -invariant symmetric bilinear forms are $\{\lambda \eta : \lambda \in \mathbb{R}\}$. For $n = 1$ the conclusion fails: the wedge $\{x \geq |t|\}$ is invariant.*

Proof. An invariant cone is a union of orbits: sheets of two-sheeted hyperboloids, null half-cones, one-sheeted hyperboloids and the origin. For $n \geq 2$ a one-sheeted hyperboloid contains v and $-v$ and is excluded by pointedness. Invariance under positive scaling, convexity and closedness leave one closed light cone. The standard representation is absolutely irreducible, so invariant symmetric forms form a line. $\square$

10 Scale and transport

Hypothesis 10.1 (H-int). Local scale representatives $g_i = \Omega_i^2 g$ on a cover differ on overlaps by positive functions $\rho_{ij} = \Omega_i/\Omega_j$ forming an exact cocycle, $\rho_{ij} = \sigma_i/\sigma_j$.

Proposition 10.2 (Scale representative). *A global metric representative of $[g]$ exists on a connected stratum iff H-int holds. A nonzero loop product of ρ is scale holonomy: a unit transported around the loop returns multiplied by it. Data along one worldline fix at most one normalization along that worldline.*

Proposition 10.3 (Strict transport). *If two representatives g and $\Omega^2 g$ are both preserved by a strict transport on a transport-connected stratum, Ω is constant there.*

Proof. $P_{qr}^*(\Omega^2 g)_r = (\Omega^2 g)_q$ and $P_{qr}^* g_r = g_q$ give $\Omega(r) = \Omega(q)$ by nondegeneracy. $\square$

Imported Theorem 10.4 (Induced transport). *On a regular embedded metric stratum with orthogonal projector P , the induced connection is $\nabla^W = P\nabla^{LC}$ and the second fundamental form is $(I - P)\nabla^{LC}$.*

No-Go 10.5 (Entropy does not select a connection). $H(X)$, $H(Q)$ and $H(X | Q)$ are invariant under fibre-preserving automorphisms, and horizontal lifts are not; no function of these entropies selects a lift.

Proposition 10.6 (Law-defined transport). *A transport defined from laws descends uniquely to Q_S (Corollary 5.4). A latent transport W descends iff $D\pi(W_x)$ is constant on fibres.*

11 Duration

Let $T_i : H \rightarrow \mathbb{R}$ be the readouts of clock protocols on a history space H , i.e. natural modules of the clock channels.

Theorem 11.1 (Clock transition groupoid). *If $\mathcal{K}_{T_i} = \mathcal{K}_{T_j}$, there is a unique bijection $f_{ji} : T_i(H) \rightarrow T_j(H)$ with $T_j = f_{ji} \circ T_i$, and*

$$f_{ii} = \text{id}, \quad f_{ij} = f_{ji}^{-1}, \quad f_{ki} = f_{kj} \circ f_{ji}.$$

If $\mathcal{K}_{T_i} \neq \mathcal{K}_{T_j}$ no such map exists.

Proof. Theorem 3.2 applied to the common quotient; uniqueness closes composition. If the kernels differ, some pair is identified by one readout and separated by the other. $\square$

Theorem 11.2 (Affine clocks). *Suppose a flow Φ_s belongs to Stab , $T_i \circ \Phi_s = T_i + a_i s$ with $a_i > 0$, and each $T_i(H)$ is an interval $(H\text{-}\Gamma)$. Then every f_{ji} is affine with slope a_j/a_i . The clocks are charts of one duration object, unique up to a positive unit and an origin on each connected component.*

Proof. $f_{ji}(T_i \circ \Phi_s) = T_j \circ \Phi_s$ gives $f_{ji}(t + a_i s) = f_{ji}(t) + a_j s$ for all admissible t, s . $\square$

Proposition 11.3 (Path independence). *An accumulated duration T is a function of a declared increment Δt iff $\mathcal{K}_{\Delta t} \subseteq \mathcal{K}_T$ on H . Unsigned elapsed duration is invariant under path reversal and fixes no time orientation.*

Imported Theorem 11.4 (Coherence duration; [16]). *For multiplicative coherence operators C , $\mathcal{T} = -\log C$ is additive on commuting families, and the defect of additivity is the Baker–Campbell–Hausdorff series $\frac{1}{2}[\mathcal{T}_1, \mathcal{T}_2] + \dots$.*

12 Physical distance

12.1 No internal unit

No-Go 12.1 (Unit). 1. If a readout scale s is not anchored, the law depends on $(x/s, \sigma/s)$, and the dilation $(x, \sigma, s) \mapsto (kx, k\sigma, ks)$ lies in Gau.

2. Markov-invariant metrics on finite simplices are c times the Fisher metric with $c > 0$ free [5].

3. The set $\{k_B, T, m\}$ contains no quantity of dimension length; a length requires an action or a pair of clock frequency and speed [4]. Thermodynamic length is invariant under a change of energy unit [6].

4. A flat metric is Euclidean in some chart; the chart need not be the physical one.

Proposition 12.2 (Counted channel removes dilation). *If F contains a counted channel whose law changes under a physical dilation, that dilation is not in Stab. If F contains only continuous channels whose relabeling groups include rescaling, dilations may lie in Stab.*

Proof. Lemma 4.4: no relabeling compensates a changed count law. □

12.2 Displacement and invariant metric

Definition 12.3 (Displacement). A displacement is an element of Γ acting on a target chart; concatenation is composition.

Imported Theorem 12.4 (Ratio scale; [9, 10]). *A monotone additive representation of concatenation is linear and unique up to one positive constant.*

Theorem 12.5 (Two-point invariant). *Suppose the latent space contains station and target positions in $\mathbb{R}^n$ and Gau contains the diagonal action of $SE(n)$, $n \geq 2$, or of $E(n)$, $n \geq 1$ (H - Γ). Then the orbit space of station–target pairs is $[0, \infty)$, parametrized by Euclidean separation ρ , and every natural two-point scalar is a function of ρ .*

Proof. The group is transitive on pairs at fixed separation: rotate (or, for $n = 1$, reflect) the difference vector, then translate. An invariant function is constant on orbits. □

Theorem 12.6 (Metric proportionality). *Suppose relative Euclidean motions of the target lie in Γ , act transitively on the target chart, and have isotropy acting irreducibly on the tangent space (H - Γ). Then every natural metric module on the chart satisfies $\bar{g} = \lambda^2 h$, where h is the Euclidean metric of ρ and $\lambda > 0$ is constant.*

Proof. $\bar{g}$ and h are Γ -invariant (Theorem 5.3). The invariant symmetric forms of an irreducible compact isotropy form a line, since the commutant is $\mathbb{R}$, $\mathbb{C}$ or $\mathbb{H}$ and its symmetric part is $\mathbb{R}$; so $\bar{g} = \lambda(q)^2 h$. The ratio λ is a Γ -invariant function and Γ is transitive. □

Imported Theorem 12.7 (Unit-distance rigidity; [2]). *For $n \geq 2$, a map $\mathbb{R}^n \rightarrow \mathbb{R}^n$ preserving distance one is an isometry. For $n = 1$ this fails: $x \mapsto x + 0.1 \sin 2\pi x$ preserves distance one and not distance one half.*

12.3 Anchor and identification

Hypothesis 12.8 (H-SI). The slice contains a two-way timing protocol that counts caesium cycles; c and $\Delta\nu_{\text{Cs}}$ take their exact SI values [21]; and round-trip counts add along concatenations for which ρ attains equality in the triangle inequality.

Proposition 12.9 (Radar distance). *Under H-SI the round-trip count n_{Cs} determines*

$$d_{\text{rad}} = \frac{c}{2\Delta\nu_{\text{Cs}}} n_{\text{Cs}},$$

and the factor $\frac{1}{2}$ is independent of the one-way synchronization convention [20]: with one-way speeds $c/(2\varepsilon)$ and $c/(2(1-\varepsilon))$ the round-trip time does not depend on ε .

Proposition 12.10 (Radar and rod distance). *On a static slice with metric representative $W^2(-dt^2 + dx^2)$ and a clock with $W = 1$ at the station, radar distance is independent of W away from the station, while rod distance is $\int W dx$. They agree on a path iff the exchanged-frequency ratio $W(x)/W(0)$ equals one along it. If translations along the path act in Γ on the clock channel, this ratio equals one.*

Proof. Null curves are conformally invariant, so the round-trip proper time at the station does not depend on W elsewhere. If a translation is in Stab , it maps the clock law at x to the clock law at $x + a$ with trivial relabeling on counts (Lemma 4.4), so the counted rates agree. $\square$

Hypothesis 12.11 (H-med). The propagation speed in the protocol is c , or is anchored by a protocol whose own anchor is exact, or the slowness field is resolved by a ray family with trivial null space.

Proposition 12.12 (Media). *Delay data are line integrals of slowness. Without H-med, thickness and speed are exchangeable (the pair (d, v) has a dilation fibre), and an incomplete ray family leaves a nonzero invisible slowness sector [14].*

Hypothesis 12.13 (H-val). The proportionality constant λ fixed by one anchor predicts a second, held-out anchor within a declared tolerance.

Theorem 12.14 (Physical distance). *Assume H-reg, H-nuis, H-SI, H-med, H-val, the Γ -conditions of Theorems 12.5 and 12.6, and translation membership of Proposition 12.10. Then the unique Γ -invariant metric h has distance d_{rad} in metres, and for every natural experiment $\mathcal{E}$ on the chart*

$$\bar{g}_{\mathcal{E}} = \lambda_{\mathcal{E}}^2 h, \quad d_{\text{phys}} = d_{\text{RG}}^{\mathcal{E}} / \lambda_{\mathcal{E}},$$

with $\lambda_{\mathcal{E}}$ constant. RG angles equal physical angles. In rank one the identification additionally requires the concatenation law of Imported Theorem 12.4.

Proof. Theorems 12.5 and 12.6 give $\bar{g}_{\mathcal{E}} = \lambda_{\mathcal{E}}^2 h$ with h determined by ρ . Radar count is a natural two-point scalar, hence a function of ρ (Theorem 12.5); by H-SI it is additive along collinear concatenations, hence linear (Imported Theorem 12.4). Proposition 12.9 fixes the constant in metres; Proposition 12.10 and H-med identify radar with rod distance; H-val excludes a response that fits one anchor only. In rank at least two Imported Theorem 12.7 makes one anchored distance sufficient. $\square$

13 The assembled theorem

Theorem 13.1 (Kernel–stabilizer theorem). *Let $\mathcal{S}$ be an observation system with joint law F , kernel groupoid $\mathcal{K}$ and effective symmetry Γ , and let all modules considered be natural. On each regular stratum:*

1. exact ignorance is $\mathcal{K}$; infinitesimal ignorance is $V = \text{rad } g_F$;
2. every module descends to the quotient and is Γ -invariant;
3. under *H-sound* and *H-coh*, completeness is constant on Γ_G -orbits;
4. clocks with equal kernels form one transition groupoid; a time flow in Stab makes it affine;
5. $SO^+(1, n) \leq \Gamma_q$, $n \geq 2$, forces the light cone and the conformal ray;
6. a counted channel excludes dilations from Γ ; Euclidean diagonal gauge and relative Euclidean symmetry force one two-point invariant and $\bar{g}_{\mathcal{E}} = \lambda_{\mathcal{E}}^2 h$;
7. under the hypotheses of Theorem 12.14, h is physical distance in metres.

Corollary 13.2 (Compatibility equations). *On every regular stratum claimed for a slice the following hold simultaneously:*

$$\text{rad } g_F = V, \quad \text{isom}(\bar{g}_{\mathcal{E}}) = \text{isom}(h), \quad \Gamma_q C_q = C_q, \quad \lambda_{\mathcal{E}} \text{ constant},$$

the clock transition groupoid is Γ -invariant, and the held-out anchor is predicted. Each equation depends on the same subgroups of Γ ; if a subgroup is absent, every equation that uses it fails.

14 The ladder

Dependencies point only to earlier rungs. Status: F forced, I imported, H hypothesis-conditioned (named), Γ Γ -conditional, N no-go, O open.

Rung	Statement	Status	Depends	Ref.
<i>Exact layer</i>				
L0	Observation system, joint law, relabeling group	input	—	2.1
L1	Kernel groupoid; $Q \cong \text{im } F$	F	L0	3.1
L2	Universal factorization; invariants are functions on Q	F	L1	3.2
L3	Joint kernel is the intersection; $Q \hookrightarrow \prod Q_i$ not onto	F	L1	3.3
L4	Garbling enlarges the kernel	F	L1	3.4
L5	Exact layer determines no higher structure	N	L2	3.5
<i>Symmetry layer</i>				
L6	$1 \rightarrow \text{Gau} \rightarrow \text{Stab} \rightarrow \Gamma \rightarrow 1$; Γ faithful on Q	F	L1	4.2
L7	The sequence need not split	N	L6	4.3
L8	Count rigidity	F	L0	4.4
L9	Orbit-quotient comparison	H:	L1	4.6
		sound, compl		
L10	Rank, separation, completeness, reconstruction are distinct	N	L2,L9	4.7
L11	Coherent completeness	H: coh	L6,L9	4.8
<i>Differential layer</i>				
L12	Tangent sequence; infinitesimal completeness defect	H: reg	L2,L9	5.2
L13	Natural descent: basic, Γ -invariant, radical = V	F	L6,L12	5.3
L14	Descent of law-defined structures is automatic	F	L13	5.4
L15	Singular strata: slices and tangent cones	I	L12	5.5
L16	First-order tangent misses one-sided directions at walls	F	L15	5.6
<i>Second-order layer</i>				
L17	Fisher is natural; $\text{rad } g_F = V$; Gaussian form	F	L13	6.2

Rung	Statement	Status	Depends	Ref.
L18	Regular reconstruction of (R, N, Σ)	I	L17	6.3
L19	Fisher does not determine the covariance	N	L18	6.4
L20	Second-order quotient has dimension $p + q + pq$	F	L2,L18	6.5
L21	Second-order shadow is not the law	N	L20	6.6
L22	Efficient metric: conventions and nuisance wall	H: nuis	L17	6.9
L23	Data processing contracts Fisher	I	L4,L17	6.10
L24	Kernel order is not Blackwell order	N	L4,L23	6.11
<i>Ignorance</i>				
L25	Persistent null; finite attainment	H: sens	L3,L17	7.2
L26	Sensor, precision and representation nulls	F	L25	7.3
L27	Finite-sample boundary reconstruction	I	L17,L22	7.4
<i>Statistical distance</i>				
L28	RG distance: representative- and chart-free, Γ -invariant, dimensionless	H: reg, nuis	L13,L22	8.2
L29	Garbling does not increase RG length	F	L23,L28	8.3
L30	No angle on rank-one strata	N	L28	8.4
L31	Finite divergences are not additive	N	L28	8.5
L32	RG distance is experiment-relative	N	L28,L29	8.6
<i>Order and cone</i>				
L33	Fibre-compatible reachability descends	F	L2	9.1
L34	Projective typing of the cone	N	L33	9.3
L35	Realization branches	H: top	L15,L34	9.4
L36	Quadratic cone determines a conformal ray	I	L35	9.5
L37	Lorentz isotropy forces the light cone ($n \geq 2$)	Γ	L6,L13,L35	9.6
<i>Scale and transport</i>				
L38	Scale representative iff exact cocycle; holonomy	H: int	L36	10.2
L39	Strict transport fixes the conformal factor	F	L38	10.3
L40	Induced connection on regular strata	I	L12	10.4
L41	Entropy does not select a connection	N	L2	10.5
L42	Law-defined transport descends; latent transport iff projectable	F / H	L14	10.6
<i>Duration</i>				
L43	Clock transition groupoid	F	L2	11.1
L44	Affine clocks under a stabilizer time flow	Γ	L6,L43	11.2
L45	Path independence as kernel inclusion; no orientation	F	L43	11.3
L46	Coherence duration and BCH memory	I	L43	11.4
<i>Physical distance</i>				
L47	No internal unit	N	L17,L28	12.1
L48	Counted channel removes dilation	F	L3,L6,L8	12.2
L49	Ratio scale	I	L6	12.4
L50	Two-point invariant	Γ	L6,L13	12.5
L51	Metric proportionality $\bar{g} = \lambda^2 h$	Γ	L13,L50	12.6
L52	Unit-distance rigidity ($n \geq 2$)	I	L51	12.7
L53	Radar distance; factor $\frac{1}{2}$	H: SI	L8,L43	12.9
L54	Radar equals rod distance iff no exchanged-frequency shift	Γ	L38,L48,L53	12.10
L55	Media: dilation fibre and invisible slowness	H: med	L25,L53	12.12
L56	Physical distance	H: reg, nuis, SI, med, val; Γ	L47–L55	12.14

Rung	Statement	Status	Depends	Ref.
<i>Assembly</i>				
L57	Kernel-stabilizer theorem; compatibility equations	Γ	L1–L56	13.1
<i>Open</i>				
L58	Canonical connection for latent transport	O	L40–L42	16.1
L59	Exhaustion of the residual characteristic branch	O	L35	16.2
L60	Non-static, quantum, non-dominated and infinite-dimensional systems	O	L57	16.3

15 Procedure

An application of RG to a physical slice consists of the following steps, each producing the object named.

1. Declare the protocols, the latent space and the relabeling group; form the joint law. (L0)
2. Determine the kernel by equality-of-law tests; obtain Q and its invariants. (L1–L3)
3. Test candidate generators of Stab by equality of laws after the declared operation and relabeling; record Γ . (L6)
4. Certify regular strata: constant rank of the law map, and of target and nuisance blocks; locate walls and use slices or jets there. (L12, L15, L22)
5. Compute natural modules: Fisher tensor, efficient metric, invariants, clock readouts, cone field. (L13, L17)
6. Read off the Γ -conditional structure: affine duration, cone type, two-point invariant, metric proportionality. (L37, L44, L50, L51)
7. Anchor: include a counted two-way timing channel; verify translation membership on the clock channel; resolve the medium. (L48, L53–L55)
8. Validate with a held-out anchor and check the compatibility equations. (L56, L57)

A failure at any step is attributed to one of: kernel misspecification; a wall; incomplete saturation; a smaller Γ ; a non-natural module; a failed anchor, medium or scale hypothesis; statistical rejection; or reliance on L58.

16 Open problems

Open Problem 16.1 (Latent connection). Construct a canonical connection on the quotient for a transport generated by latent dynamics, from the response Jacobian, the covariance and a memory kernel. Entropy does not select one (No-Go [10.5](#)).

Open Problem 16.2 (Residual branch). Classify the characteristic geometries of Theorem [9.4](#)(5).

Open Problem 16.3 (Extensions). Extend Theorems [12.14](#) and [13.1](#) to non-static slices (where translations or rotations are absent from Γ and radar and rod distance differ), to quantum protocols, and to non-dominated or infinite-dimensional experiments.

Remark 16.4 (Computational witnesses). Each No-Go and each finite example in this paper has a deterministic numerical witness in the accompanying scripts. The witnesses check the stated counterexamples; they are not premises of any proof.

References

- [1] S. Amari, H. Nagaoka, *Methods of Information Geometry*, AMS/Oxford, 2000.
- [2] F. S. Beckman, D. A. Quarles, On isometries of Euclidean spaces, *Proc. AMS* 4 (1953), 810–815.
- [3] D. Blackwell, Comparison of experiments, *Proc. 2nd Berkeley Symp.*, 1951.
- [4] E. Buckingham, On physically similar systems, *Phys. Rev.* 4 (1914), 345–376.
- [5] N. N. Čencov, *Statistical Decision Rules and Optimal Inference*, AMS, 1982.
- [6] G. E. Crooks, Measuring thermodynamic length, *Phys. Rev. Lett.* 99 (2007), 100602.
- [7] M. Goresky, R. MacPherson, *Stratified Morse Theory*, Springer, 1988.
- [8] S. W. Hawking, A. R. King, P. J. McCarthy, A new topology for curved space–time which incorporates the causal, differential, and conformal structures, *J. Math. Phys.* 17 (1976), 174–181.
- [9] O. Hölder, Die Axiome der Quantität und die Lehre vom Mass, *Ber. Sächs. Ges. Wiss.* 53 (1901), 1–64.
- [10] D. H. Krantz, R. D. Luce, P. Suppes, A. Tversky, *Foundations of Measurement, Vol. I*, Academic Press, 1971.
- [11] L. Le Cam, *Comparison of Experiments*, IMS, 1996.
- [12] D. Luna, Slices étales, *Bull. Soc. Math. France, Mémoire* 33 (1973), 81–105.
- [13] D. B. Malament, The class of continuous timelike curves determines the topology of spacetime, *J. Math. Phys.* 18 (1977), 1399–1404.
- [14] L. Pestov, G. Uhlmann, Two dimensional compact simple Riemannian manifolds are boundary distance rigid, *Ann. Math.* 161 (2005), 1093–1110.
- [15] M. J. Pflaum, H. Posthuma, X. Tang, Geometry of orbit spaces of proper Lie groupoids, *J. reine angew. Math.* 694 (2014), 49–84.
- [16] M. Rada, *Operational Duration*, 2026.
- [17] M. Rada, *The Law and Boundary of Observational Ignorance*, 2026.
- [18] M. Rada, *Observation Laws, Universal Quotients, and Saturated Observable Geometry*, 2026.
- [19] M. Rada, *Resolution Geometry: Reconstruction and Identifiability*, 2026.
- [20] H. Reichenbach, *Philosophie der Raum-Zeit-Lehre*, de Gruyter, 1928.
- [21] BIPM, *The International System of Units (SI)*, 9th ed., 2019.
- [22] F. Zhang (ed.), *The Schur Complement and Its Applications*, Springer, 2005.
- [23] N. T. Zung, Proper groupoids and momentum maps: linearization, affinity, and convexity, *Ann. Sci. Éc. Norm. Supér.* 39 (2006), 841–869.